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TATA POWER-DDL
Towards a Greener Tomorrow

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India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

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Domain Language Model for Power & Energy

Sankar Patnaik , Director, Citi

Digital Colleagues in Action

What Do Digital Colleagues Do?

AI Research Assistants
  Analyze millions of documents in seconds
AI Data Analysts
  Detect patterns and risks automatically
Autonomous Robots
  Inspect infrastructure and hazardous environments
AI Predictive Systems
  Prevent failures before they occur

AI is not replacing workers.

- It is **augmenting human capabilities**.

Domain Language Model

Scenario

6:30 PM — Peak demand in the city
It's evening.Homes switch on lights, air conditioners, elevators, and appliances.
Factories are still running.

- Electricity demand suddenly rises.
- Inside the power grid control room, engineers monitor hundred s of screens.

Suddenly…

- ⚠️ Alarm! Transformer T3 has tripped and part of the city may lose power.

Without AI

The engineer must investigate manually.

He starts checking multiple systems:

- SCADA logs to detect the fault
- Load history to see demand spikes
- Maintenance records of the transformer
- Weather data for heat impact

Each system is separate.
Each analysis takes time.
⏱️ Total response time: 30–60 minutes
Meanwhile thousands of homes could lose electricity.

With Domain Language Model (AI)

The engineer simply asks the system:
"Why did Transformer T3 trip?"

The AI instantly analyzes:

- Grid load
- Sensor data
- Maintenance history
- Weather conditions
- Historical failures

AI responds within seconds:
⚡ "Transformer T3 exceeded 95% capacity due to industrial demand surge. Recommend shifting 15 MW load to Substation B."

Result

Power is restored **before customers even notice a problem.**

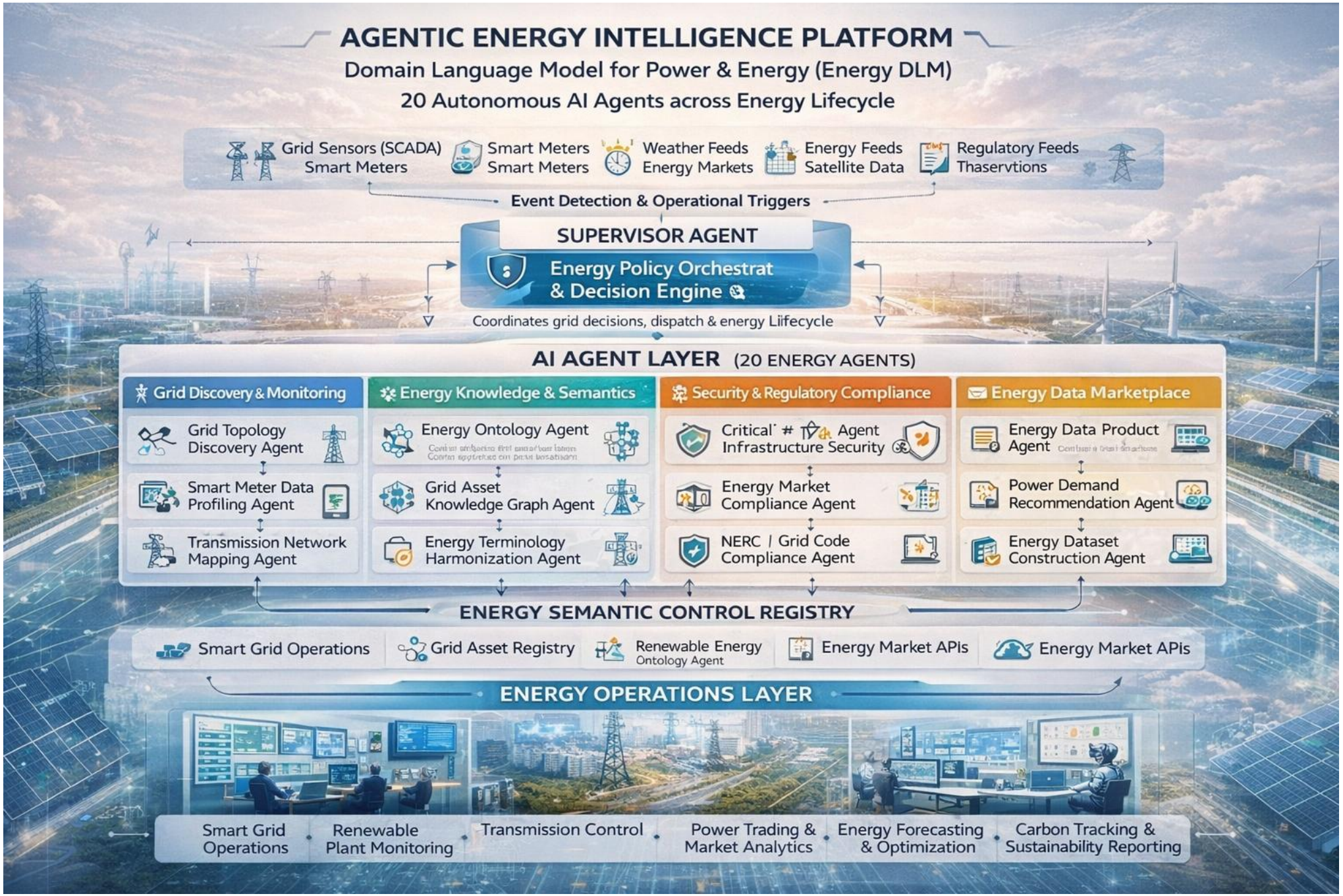
What is DLM (Domain Language Model)

A Domain Language Model is like an AI engineer who has studied only one industry deeply.

- Instead of knowing a little about everything like ChatGPT, it knows a lot about one domain — such as electricity, power grids, substations, load demand, and outages.

Think of it like this:

- A general doctor knows many diseases.
- A cardiologist specializes in heart problems.
- A Domain Language Model is the cardiologist of AI.


AGENTIC ENERGY INTELLIGENCE PLATFORM
Domain Language Model for Power & Energy (Energy DLM)
20 Autonomous AI Agents across Energy Lifecycle
Grid Sensors (SCADA)
Smart Meters
Smart Meters
Smart Meters
Weather Feeds
Energy Markets
Energy Feeds
Satellite Data
Regulatory Feeds
Thaservtions
Event Detection & Operational Triggers
SUPERVISOR AGENT
Energy Policy Orchestrat
& Decision Engine
Coordinates grid decisions, dispatch & energy Liifecycle
AI AGENT LAYER (20 ENERGY AGENTS)
Grid Discovery & Monitoring
Grid Topology Discovery Agent
Smart Meter Data Profiling Agent
Transmission Network Mapping Agent
Energy Knowledge & Semantics
Energy Ontology Agent
Grid Asset Knowledge Graph Agent
Energy Terminology Harmonization Agent
Security & Regulatory Compliance
Agent
Infrastructure Security
Energy Market Compliance Agent
NERC / Grid Code Compliance Agent
Energy Data Marketplace
Energy Data Product Agent
Power Demand Recommendation Agent
Energy Dataset Construction Agent
ENERGY SEMANTIC CONTROL REGISTRY
Smart Grid Operations
Grid Asset Registry
Renewable Energy Ontology Agent
Energy Market APIs
Energy Market APIs
ENERGY OPERATIONS LAYER
Smart Grid Operations
Renewable Plant Monitoring
Transmission Control
Power Trading & Market Analytics
Energy Forecasting & Optimization
Carbon Tracking & Sustainability Reporting

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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